Jiahao Song

LinkedIn   Google Scholar   jis090@ucsd.edu   (858) 214-6592

RESEARCH INTERESTS

Development of energy-efficient circuits and systems for artificial intelligence (AI) and combinatorial optimization.

- Analog and Digital Compute-in-Memory Circuits for Artificial Intelligence Applications
- Energy-Efficient Circuits and Accelerators for Combinatorial Optimization Problems
- Design Automation for Mixed-Signal and Compute-in-Memory Circuits

APPOINTMENTS

Postdoctoral Research Fellow, University of California San Diego — Jul. 2024 – Present
Advisor: Prof. Gert Cauwenberghs

Postdoctoral Research Fellow, Peking University — Jul. 2022 – Jun. 2024
Advisor: Prof. Xiyuan Tang, Runsheng Wang, and Yuan Wang

EDUCATION

Ph.D. in Microelectronics
Peking University, Beijing, China — 2017 – 2022
Advisor: Prof. Runsheng Wang, Yuan Wang and Xiyuan Tang

B.S. in Electronic Science and Technology (Hons.)
University of Electronic Science and Technology of China, Chenghu, China — 2013 – 2017

SELECTED PUBLICATIONS

[1] X. Qiao, Y. Yang, **Jiahao Song**, et al., "MixCIM: A Hybrid Computing-in-Memory Macro With Less Data-Movement and Better Memory-Reuse for Depthwise Separable Neural Networks," IEEE Journal of Solid-State Circuits (JSSC), 2026.

[2] **Jiahao Song**, et al., "A 1131-Kb/mm^2 14.0-to-53.3-TOPS/W 8-Bit Analog-Assisted Digital Compute-in-Memory With Hybrid Local-Refresh eDRAM for Attention Computing," IEEE Journal of Solid-State Circuits (JSSC), 2025.

[3] B. Xu*, **Jiahao Song***, et al., "A 0.2-V Edge-Pursuit Continuous-Time Ising Machine Using Ring-Oscillator Collapse and Delay-Enabled Positive Feedback," IEEE Journal of Solid-State Circuits (JSSC), 2025. (* co-first author)

[4] **Jiahao Song**, et al., "A 4-Bit Calibration-Free Computing-In-Memory Macro with 3T1C Current-Programmed Dynamic-Cascode Multi-Level-Cell eDRAM," IEEE Journal of Solid-State Circuits (JSSC), 2024.

[5] **Jiahao Song**, et al., "A 28nm 16Kb Bit-Scalable Charge-Domain SRAM Compute-in-Memory Macro," IEEE Transactions on Circuits and Systems I: Regular Papers (TCAS-I), 2023.

[6] **Jiahao Song**, et al., "A 3T eDRAM In-Memory Physically Unclonable Function With Spatial Majority Voting Stabilization," IEEE Solid-State Circuits Letters (SSC-L), 2022.

[7] **Jiahao Song**, et al., "TD-SRAM: Time-Domain In-Memory Computing for Binary Neural Networks," IEEE Transactions on Circuits and Systems I: Regular Papers (TCAS-I), 2021. (**Highlight of 2021 August Issue**)

[8] H. Diao, H. Luo, **Jiahao Song**, et al., “A Computing-in-Memory Engine Supporting One-Shot Floating-Point NN Inference and On-Device Fine-Tuning for Edge AI,” IEEE Journal of Solid-State Circuits (JSSC), 2025.

[9] Z. Wu, **Jiahao Song**, et al., “A Variation-Tolerant Continuous-Time Ising Machine With eDRAM-Based Spin Interaction and Leaked Negative Feedback Annealing,” IEEE Journal of Solid-State Circuits (JSSC), 2024.

CONFERENCE PUBLICATIONS (SELECTED)

[1] **Jiahao Song***, Z. Wu*, et al., “A Variation-Tolerant In-eDRAM Continuous-Time Ising Machine Featuring 15-Level Coefficients and Leaked Negative-Feedback Annealing,” IEEE International Solid-State Circuits Conference (ISSCC), 2024. (* co-first author)

[2] X. Qiao* **Jiahao Song***, et al., “MixCIM: A Hybrid-Cell-Based Computing-in-Memory Macro with Less-Data-Movement and Activation-Memory-Reuse for Depthwise Separable Neural Networks,” IEEE Custom Integrated Circuits Conference (CICC), 2024. (* co-first author)

[3] **Jiahao Song**, et al., “A Calibration-Free 15-level/Cell eDRAM Computing-in-Memory Macro with 3T1C Current-Programmed Dynamic-Cascoded MLC achieving 233-to-304-TOPS/W 4b MAC,” IEEE Custom Integrated Circuits Conference (CICC), 2023.

[4] **Jiahao Song**, et al., “Spike-CIM: A 290TOPS/W Spike-Encoding Sparsity-Adaptive Computing-in-Memory Macro with Differential Charge-Domain Integrate-and-Fire,” IEEE Asian Solid-State Circuits Conference (ASSCC), 2022.

[5] **Jiahao Song**, et al., “A 16Kb Transpose 6T SRAM In-Memory-Computing Macro based on Robust Charge-Domain Computing,” IEEE Asian Solid-State Circuits Conference (ASSCC), 2021.

[6] B. Xu, J. Ruan, **Jiahao Song**, et al., “A 28-nm CIM-Based Annealing Processor with One-Shot Weight Access, Dynamic Batching, and Bubble-Less Pipeline for Large-Scale Path Planning Problems,” IEEE Custom Integrated Circuits Conference (CICC), 2026.

[7] M. Lee, Z. Liu, A. Uppal, **Jiahao Song**, et al., “In-Ear EEG Auditory Neurofeedback Towards Unobtrusive Sleep Enhancement,” IEEE Custom Integrated Circuits Conference (CICC), 2025.

[8] H. Zhang, **Jiahao Song**, et al., “EasyACIM: An End-to-End Automated Analog CIM with Synthesizable Architecture and Agile Design Space Exploration,” IEEE/ACM Design Automation Conference (DAC), 2024.

[9] Z. Sun, Y. Cai, **Jiahao Song**, et al., “Comprehensive Study of NBTI and Off-State Reliabilty in Sub-20 nm DRAM Technology: Trap Identification, Compact Aging Model, and Impact on Retention Degradation,” IEEE International Electron Devices Meeting (IEDM), 2024.

[10] D. Diao, H. Luo, **Jiahao Song**, et al., “A 28nm 128TFLOPS/W Computing-In-Memory Engine Supporting One-Shot Floating-Point NN Inference and On-Device Fine-Tuning for Edge AI,” IEEE Custom Integrated Circuits Conference (CICC), 2024.

[11] H. Zhang, X. Gao, H. Luo, **Jiahao Song**, et al., “SAGERoute: Synergistic Analog Routing Considering Geometric and Electrical Constraints with Manual Design Compatibility,” IEEE/ACM Design, Automation & Test in Europe (DATE), 2023. (**(Best Paper Award)**)

[12] J. Luo, W. Xu, Y. Du, B. Fu, **Jiahao Song**, et al., “Energy-and Area-efficient Fe-FinFET-based Time-Domain Mixed-Signal Computing in Memory for Edge Machine Learning,” IEEE International Electron Devices Meeting (IEDM), 2021.

[13] C. Meng, Z. Xiang, N. Liu, Y. Hu, **Jiahao Song**, et al., “DALTA: A Decomposition-Based Approximate Lookup Table Architecture,” IEEE/ACM International Conference on Computer Aided Design (ICCAD), 2021.

RESEARCH EXPERIENCE

University of California San Diego Jul. 2024 – Present

Postdoctoral Research Fellow

3D-Integrated Photonic–Electronic AI Accelerator

- Co-designed a hybrid 3D photonic–electronic neural-network accelerator integrating a high-throughput photonic front-end with an RRAM-based compute-in-memory (CIM) backend, enabling energy-efficient AI acceleration through heterogeneous computing across optical and electronic domains
- Led the circuit design of a 12nm 1T1R RRAM-based analog compute core in close collaboration with the device team, improving compute density and energy efficiency for edge AI applications
- Investigated advanced 3D integration strategies including TSV implementation and DBI-based die bonding, addressing interconnect bandwidth, integration density, and heterogeneous chip integration challenges for large-scale AI systems

Peking University Sept. 2017 – Jun. 2024

Graduate Researcher and Postdoctoral Research Fellow

Analog Compute-in-Memory Circuits for Artificial Intelligence Applications

- Developed a calibration-free current-programmed multi-level eDRAM compute-in-memory architecture, substantially improving compute accuracy and robustness against transistor-level variations while achieving over 300 TOPS/W energy efficiency.
- Developed a time-domain SRAM compute-in-memory macro with a dual-edge single-input computing topology, significantly improving energy efficiency and reducing area overhead for binary neural-network acceleration.
- Developed a calibration-free current-programmed multi-level eDRAM compute-in-memory architecture, substantially improving compute accuracy and robustness against transistor-level variations while achieving over 300 TOPS/W energy efficiency.

Digital Compute-in-Memory Circuits for Artificial Intelligence Applications

- Developed an analog-assisted digital compute-in-memory architecture for Transformer attention computing, significantly reducing 8-bit digital MAC operations through low-precision analog pre-computation while improving energy efficiency up to 53.3 TOPS/W.
- Developed a one-shot floating-point compute-in-memory engine with on-device fine-tuning capability, substantially improving floating-point AI inference throughput energy efficiency for edge AI applications.

Energy-Efficient Circuits and Accelerators for Combinatorial Optimization Problems

- Developed a variation-tolerant continuous-time Ising machine with eDRAM-based spin interactions and leaked negative-feedback annealing, enabling nanosecond-scale combinatorial optimization with 15-level coefficients and energy consumption as low as 0.33 nJ.
- Developed an edge-pursuit continuous-time Ising machine based on even-stage ring oscillators and delay-enabled positive feedback, improving solution quality and PVT robustness while achieving ultra-low-voltage operation at 0.2 V with only 12 fJ/spin energy consumption.
- eveloped a complete SRAM-based K-SAT solver with dual-path processing-in-memory architecture and incremental backtracking, achieving 100% solvability with 952× speedup and 3.4×10^5 energy reduction over software SAT solvers.

Design Automation for Mixed-Signal and Compute-in-Memory Circuits

- Developed an end-to-end automated analog compute-in-memory (ACIM) design framework with automated design-space exploration, enabling automatic generation of Pareto-optimal CIM architectures while optimizing area, throughput, and energy-efficiency trade-offs across diverse AI applications.

- Developed a synergistic analog routing framework considering both geometric and electrical constraints, enabling automated routing with EM/IR-drop-aware wire sizing and symmetry-aware optimization while achieving silicon-compatible performance on real-world taped-out analog and mixed-signal designs.

HONORS AND AWARDS

- Best Paper Award, IEEE/ACM Design, Automation and Test in Europe (DATE), 2023
- Outstanding Postdoctoral Researcher, Peking University, 2023
- Merit Student, Peking University, 2021
- Outstanding Graduate Award, Sichuan Province, China, 2017
- National Undergraduate Electronic Design Contest, Second Prize, 2016
- National Undergraduate Electronic Design Contest, Second Prize, 2015

PROFESSIONAL SERVICE

- Reviewer for leading IEEE journals, including:
  - IEEE Journal of Solid-State Circuits (JSSC)
  - IEEE Transactions on Circuits and Systems I & II (TCAS-I, TCAS-II)
  - IEEE Transactions on Very Large Scale Integration Systems (TVLSI)
  - IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)
  - IEEE Transactions on Biomedical Circuits and Systems (TBioCAS)
  - IEEE Journal of Exploratory Solid-State Computational Devices and Circuits (JxCDC)
- Reviewer for international conferences, including:
  - IEEE International Symposium on Circuits and Systems (ISCAS)
- Reviewer for additional journals, including:
  - Microelectronics Journal
  - Integration
  - Scientific Reports (Nature Portfolio)